



Pragmatism as sociotechnical fallibilism and meliorism

Henrik Rydenfelt¹

Received: 23 February 2025 / Accepted: 15 July 2025 / Published online: 21 July 2025
© The Author(s) 2025

Abstract

Watson, Mökander, and Floridi (Watson et al., AI Soc, 2024) propose that contemporary discussions in AI ethics revolve around two competing narratives, which they term sociotechnical dogmatism and sociotechnical skepticism. As an alternative to this dichotomy, they introduce sociotechnical pragmatism, which they describe as a constructive synthesis of these opposing views. However, the distinction they draw between these narratives conflates their epistemological and ethical dimensions, making it difficult to classify possible alternative positions. As a result, sociotechnical pragmatism, as formulated by Watson et al., does not fully account for the potential of pragmatism as an approach to inquiry, regulation, and governance of technology. Rather than treating regulation as a reactive measure, pragmatism seeks to integrate inquiry into the development process itself, ensuring that both the consequences of new technologies and their ethical implications are critically examined in advance. In this manner, pragmatism as sociotechnical fallibilism and meliorism provides an alternative to a large-scale societal experimentation with technology which treats harm as an inevitable cost of progress.

Keywords Artificial intelligence · Machine learning · Ethics · AI ethics · Epistemology · Pragmatism

1 Introduction

Debates in the ethics of technology have been shaped by diverging portrayals of technological development, with some advocating for rapid progress as a force for societal good and others emphasizing the risks, unintended consequences, and ethical challenges that emerging technologies present. David S. Watson, Jakob Mökander, and Luciano Floridi (2024) propose that contemporary discussions in AI ethics are structured around two competing narratives, which they term sociotechnical dogmatism and sociotechnical skepticism. With a broad audience in mind—including AI developers, the public, and policymakers—the narratives do not engage directly with specific issues that have been central to academic discussions in AI ethics, such as bias and discrimination, privacy and surveillance, or autonomy and human agency. Rather, they are presented as broad perspectives on ongoing sociotechnical developments, including the continued advancement and deployment of AI, and their wider ethical, social and societal consequences. The first narrative focuses on technological progress and

the potential for AI systems to generate positive social and societal outcomes. It highlights the advantages of continued AI development, arguing that these technologies can drive innovation and address complex problems. The second narrative, in contrast, centers on the risks and harms AI technologies have caused and may continue to cause, advocating for increased oversight and regulation. Watson et al. argue that while both narratives capture relevant aspects of the broader discourse on AI ethics, each remains incomplete: sociotechnical dogmatism fails to account for the unjust consequences of technological progress, while sociotechnical skepticism risks rejecting beneficial technological developments. To move beyond this dichotomy, they introduce sociotechnical pragmatism as an alternative perspective, describing it as a constructive synthesis of these opposing narratives, characterized by empirical inquiry, democratic deliberation, and regulatory and policy interventions that take into account both the potential benefits and risks of AI technologies.

Seeking a mediating position between these alternatives from within the tradition of philosophical pragmatism is a welcome and well-motivated contribution. However, it is argued that the distinction between the two dominant narratives conflates their epistemological and ethical dimensions, making it difficult to clearly delineate alternative positions. As a result, sociotechnical pragmatism, as outlined

✉ Henrik Rydenfelt
henrik.rydenfelt@gmail.com

¹ University of Helsinki, Helsinki, Finland

by Watson et al., does not fully capture the potential of pragmatism as an approach to sociotechnical issues, including technological inquiry, regulation and governance. It is argued that the contribution of pragmatism—and the middle path it offers—can be understood through the interplay of two central notions. The first is fallibilism, which holds that any of our beliefs may be mistaken, rejecting both dogmatism and skepticism. The second is meliorism, which seeks a middle path between optimism and pessimism, proposing that improvement is possible but neither inevitable nor guaranteed. To set the stage for this argument, the next section presents Watson et al.'s vision of sociotechnical pragmatism as an alternative to both dogmatic and skeptical approaches to sociotechnical change. The third section introduces pragmatism as a philosophical tradition, focusing on its perspectives on science and ethics, and elaborating the concepts of fallibilism and meliorism. The fourth section engages in a critical comparison between Watson et al.'s proposal and the present account of pragmatism as sociotechnical fallibilism and meliorism, highlighting their conceptual differences. Finally, the fifth and final section clarifies the practical difference between Watson et al.'s proposal and the conception of pragmatism as sociotechnical fallibilism and meliorism by considering how the latter can address the challenge posed by the rapid introduction of technologies such as AI, often accompanied by speculative promises and the risk of significant harm.

2 Three narratives of sociotechnical development

Watson et al. distinguish between two competing narratives in AI ethics, emerging in response to rapid advances in machine learning and the growing adoption of AI systems. Although their account is presented as a review of AI ethics discourse, it is less concerned with particular ethical challenges or approaches and more oriented toward overarching narratives of sociotechnical change. While they acknowledge key concerns in the AI ethics literature—such as bias, privacy, and fears surrounding artificial general intelligence (AGI)—as well as the range of normative ethical perspectives adopted by researchers, they focus on overarching sociotechnical trajectories articulated in a wider body of literature, including policy documents, journalistic accounts, and public statements by political and corporate actors. Within this framing, they identify two dominant narratives, the first of which they term *sociotechnical dogmatism*, which centers on the technological progress and potential of AI systems to generate positive social and societal outcomes. This narrative underscores the benefits of continued development and deployment of AI, arguing that these technologies can drive innovation and solve complex societal problems. The

competing narrative, *sociotechnical skepticism*, focuses on the harms that technologies have caused (and may continue to cause) to individuals, groups, and the environment; it calls for increased regulation and oversight of AI systems to address their potential harms. This narrative takes a critical stance on new technologies and questions whether AI can truly provide solutions to human problems. These two positions are not tied to specific individuals or groups but are instead presented as ideal types in contemporary discourse on AI ethics. However, they connect sociotechnical dogmatism with proponents in Silicon Valley entrepreneurs, business executives, and the policymakers and researchers who advocate for the rapid advancement of AI technologies, and associate sociotechnical skepticism with critical research on technology, including datafication, algorithms, and AI.

Watson et al. discuss topics in AI ethics where these two narratives lead to contrasting conclusions, in particular issues of fairness and justice as well as transparency and explainability in the context of these technologies and their application. They argue that both narratives are not so much wrong as incomplete. While sociotechnical dogmatism is blind to the unjust consequences and means of technological progress, the narrative of sociotechnical skepticism leads to overly broad opposition to technological advances that could contribute to improvement and progress. Their alternative, which they call *sociotechnical pragmatism*, proposes a “constructive synthesis” of the dogmatist and skeptical narratives (Watson et al. 2024, 6). Drawing on the philosophical tradition of pragmatism, this alternative promotes research, policy programs, practical solutions, and democratic deliberation. In their characterization, pragmatism maintains that theories should be judged by their success in real-world situations, and emphasizes the importance of the reciprocal relationship which exists between social and technical systems (including the history of their development); they acknowledge that no single model can provide conclusive guidance for how to evaluate particular technologies or design policies.

The salience of two alternative narratives surrounding technological developments is difficult to deny. Indeed, it may be that their emergence is not connected with the dawn of the Industrial Revolution, as Watson et al. suggest, but traced far beyond the modern era. Although no specific group or individual fully exemplifies either narrative, conflicting visions over technology and its “advances” may have been present in some shape or form as long as technology—or humanity—has existed; already Plato's Socrates recounts a “tale of the ancients” regarding the Egyptian king Thamus's criticism of the invention of writing as detrimental to memory (*Phaedrus* 274c–275c). In recent years, critical accounts have risen particularly in connection to algorithmic technologies, datafication, data collection, and AI, and their potential implications for human life—including concerns

over privacy loss and manipulation, displacement of human decision-making, creativity and thinking, risks of unfair or unjust consequences, and so on. Simultaneously, competing voices have hailed these technologies—AI in particular—as potential solutions for nearly all of humanity’s problems.

However, subsuming, under two grand narratives, a variety of critical and “skeptical” perspectives as well as more relaxed and optimistic attitudes toward technology, comes at a cost. This is already evident in the terminology employed by Watson et al. While they argue that their terms of choice—sociotechnical dogmatism and skepticism—highlight important links between debates in AI ethics and classical issues in ethics and epistemology, this choice easily appears too simplistic and problematic. This is most evident when considering their characterization of these narratives through four theses they adapt from Danaher (2022):

1. “The fact premise, which states that it is possible to assemble relevant facts;
2. The value premise, which states that there are value criteria according to which the facts can be evaluated;
3. The evaluation premise, which states that the evaluation of the facts in light of the value criteria will be positive on balance;
4. The technological premise, which states that technology plays a crucial role in ensuring that positive evaluation of the facts holds up in light of the values.” (Watson et al. 2024, 5)

Their motivation for using the labels of sociotechnical dogmatism and skepticism for the alternative narratives is partly based on the fact that they exhibit different stances toward these propositions. While sociotechnical dogmatism accepts these four premises unconditionally, in Watson et al.’s characterization, sociotechnical skepticism, in its various forms, denies the truth of some or all of them.

However, sociotechnical skepticism as discussed by Watson et al. is associated with the view that the consequences of adopting new technologies are detrimental, as well as related calls for increased regulation over their use and development. Such views objecting primarily to the third and fourth premises are, of course, ample: there is no shortage of “critical” accounts focused on the problems incurred by new technologies, having in recent decades been directed at radio, television, mass media, and now algorithmic technologies and AI. Epistemologically speaking, however, these are not skeptical views. Instead, they may amount to a full-blown denial of the value of technological developments and the availability of new technologies; in some cases, these can be stances just as “dogmatic” as (in Watson et al.’s terms) as the sociotechnical dogmatism that they oppose.

Moreover, Watson et al. situate approaches such as the sociology of scientific knowledge—which they characterize

as inspired by thinkers such as Michel Foucault and Bruno Latour—under the same skeptical banner. However, such lines of inquiry seldom take an explicit stance on the four premises delineated above. Rather, they tend to focus on the question of how particular developments, such as scientific theories, views of central social problems to be addressed by society, visions, and imaginaries concerning technological “solutions”, and so on, have come to exist and dominate, for example, following a Foucauldian genealogical approach. Such accounts are not literally skeptical—genealogical accounts, say, must stand on some “facts”. Instead, they are largely disinterested with respect to producing the kind of evaluative views concerning technologies that the four theses could circumscribe.

Indeed, the epistemological position commonly called skepticism is what Watson et al. refer to as “strong skepticism”, which challenges the first and second theses: skeptics maintain that we do not have access to the (relevant) facts or values. However, genuine skepticism results in uncertainty over the possibility of reaching any solid grounds for assessing the third and fourth theses. As Watson et al. themselves note, views of this kind leave little normative ground for policy interventions. Including such approaches within the same narrative that is connected with the demand for increased regulation results in a convoluted account of the alternatives actually at hand.

Accordingly, we can arrive at a more refined account of the narratives in question, as well as of the potential of the pragmatism alternative, by drawing a clearer distinction between the first and second theses—views on which mark the divide between different types of dogmatism and skepticism—and the third and fourth theses, which represent another kind of divide. Indeed, this will be taken up in the following section by distinguishing two central tenets developed within the pragmatist philosophical tradition.

As the subsequent discussion hopes to demonstrate, this is not a mere exercise in conceptual clarity. It helps to understand the alternative pragmatism may offer in relation to the narratives Watson et al. distinguish and the different theses that could underlie such alternatives. While Watson et al. describe sociotechnical pragmatism as an intermediary between a clearly affirmative and clearly negative response to the third and fourth theses, their view regarding how pragmatism relates to the first and second premises is more ambiguous. In connection with the first thesis, they argue that pragmatists accept that “facts are not subjective but relational” (Watson et al. 2024, 14) and that pragmatism “replaces objectivity with intersubjectivity” (*ibid.*, 15). For Watson et al., pragmatism suggests that knowledge is always contextual and qualified in this sense. The second, value-based premise receives less attention. They suggest that “what counts as morally acceptable is a revisable cultural inheritance” (*ibid.*, 7) and that pragmatism does not expect

“theoretical closure” regarding normative issues (*ibid.*, 15). However, as will be argued, it is the interplay between the four premises that sustains the promise of—and strengthens the case for—pragmatism as a viable alternative to the other narratives. The key to this potential lies in two central concepts that Watson, Mökander and Floridi leave unused and unexamined.

3 Pragmatism in science and ethics

The two concepts left unused are *fallibilism* and *meliorism*. To fully appreciate their relevance to both the pragmatist tradition and its potential as an alternative approach in AI ethics and to technology in general, however, some background is required. Pragmatism, typically seen as originating with the work of the American thinkers Charles S. Peirce, William James, and John Dewey, provides an understanding of scientific inquiry and its connections with ethical, societal, and technological issues that has gradually become more pervasive in philosophical debates. Around the turn of the twenty-first century, pragmatism experienced a resurgence across numerous debates in philosophy; in part this was because it remained unencumbered by many assumptions of early twentieth-century positivism, including reductionism about meaning and normativity—assumptions that were gradually overcome by later twentieth-century philosophy (Rydenfelt 2022a). Often viewed as mediating between philosophical extremes, pragmatism has more recently regained new ground and attention in various areas of inquiry, including ethical and societal issues.

The origins of pragmatism are commonly traced to Peirce’s maxim of pragmatism, a principled test for the meaning (and meaningfulness) of theoretical judgments. In its early formulation (though without the pragmatist label), Peirce (1878) presented a maxim for the clarification of the meaning of terms and sentences. To clarify the meanings of theoretical notions, we are to consider their conceivable effects in conduct. Later, Peirce (1903) formulated his pragmatism as the thesis that the meaning of any theoretical judgment expressed in the indicative mood can be traced in terms of the practical maxims that the acceptance of that sentence enforces. Advocating for holism concerning meaning, Peirce and the other pragmatists maintained that the practical consequences of particular beliefs or judgements must be examined within the wider context of our other commitments.

The pragmatists approached questions of epistemology and philosophy of science primarily through the central concept of inquiry. They rejected both Cartesian skepticism and the ensuing search for indubitable premises or a priori foundations for science and knowledge, the pursuit Dewey (1929) aptly called the “quest for certainty”. Instead, pragmatists argued that our inquiries must begin with what we

do not presently doubt while recognizing that any of our commitments may ultimately be subjected to doubt and critical scrutiny. Empirical inquiry is characterized by fallibilism—one concept among many coined by Peirce—which maintains that any of our views may be mistaken (Rydenfelt 2013; 2021a; 2023b).¹

Contrary to long-standing misconceptions and misrepresentations, pragmatism does not entail relativism or subjectivism, a dismissal of truth or knowledge, a disregard for theoretical research, or a crude instrumentalism that conflates or supplants truth or goodness with efficiency or “pragmatic” cash value. Peirce, James, and Dewey all argued that truth is agreement with reality, each providing a slightly differing account of how such an agreement is to be understood within the practical terms of empirical and experimental inquiry. For example, contending that truth, as the “ordinary epistemologist” would maintain, is agreement with reality, James argued that the pragmatist “insists on being more concrete and asks what such ‘agreement’ may mean in detail”, and suggested that true ideas “must point to or lead toward that reality and no other, and then the pointings and leadings must yield satisfaction as their result” (James 1909, 191). These vague formulations were not intended to imply that truth is simply “what works” as opposed to “correspondence to reality”; indeed, the idea that there is a separate pragmatist “theory” of truth is largely Bertrand Russell’s concoction (Russell 1910). Instead, realizing that the practical markers of truth, such as predictive success, vary significantly across different fields of inquiry, James and Dewey deliberately framed their ideas in flexible terms.

Resisting reductionism and a narrow conception of empirical inquiry, the pragmatists advocated for an empirical approach to normative questions, resisting both an a priori approach to ethics and the reduction of normativity to descriptive issues in the social sciences, psychology, or similar fields. They argued that although normative and descriptive inquiries address distinct problems and questions, both are amenable to a single scientific methodology. Peirce (1903) proposed examining normative issues within a triad of normative sciences: logic, ethics, and esthetics. Dewey, particularly in his writings of the 1920s and 1930s on ethics, logic, epistemology, and the philosophy of science, provided the most comprehensive exposition of an empirical and experimental approach to ethics. He argued that values can be revised, and ethical norms subjected to empirical confirmation based on their observable consequences. In keeping with general scientific inquiry, such empirical inquiry

¹ Giving a precise formulation of fallibilism faces a well-known problem with necessary truths: if there are necessary truths, we cannot be mistaken about them in the sense that our belief in a necessary truth could turn out to be false. For discussion and a disjunctive formulation, see (Rydenfelt 2021a).

into value is to take place within and through a community (Putnam 1992; Rydenfelt 2022b; 2025).

Dewey extended this view to social philosophy in terms of his conception of democracy. He was particularly concerned with the development of social policy in an era where social management and administration often pursued unexamined, predetermined goals, while the social sciences tended to eschew evaluations, leading to purely descriptive results (Dewey 1938, 487–489). Dewey advocated for transforming the change and revision of social and societal practices into more deliberate modes of social inquiry, conducted within and through a community. Such inquiry involves systematic empirical investigation into policies and practices, engaging the public as a source of information and data on problems, potential solutions, and the observed consequences of new policies (Dewey 1927; Rydenfelt 2023a; 2024). Dewey's vision has often been aligned with deliberative theories of democracy—viz., identifying democratic processes with informed, reasoned discussion and debate to solve shared problems (e.g., Bohman 2010). However, this alignment risks obscuring the experimental nature of Dewey's conception of social inquiry and democracy (Rydenfelt 2019). Dewey's vision contrasts with deliberative approaches that rest on just discussion and debate. Dewey (1935) maintained that just as scientific discovery relies on experimental observation guided by hypotheses rather than discussion alone, democratic societies should adopt the scientific method to address social issues—including normative ones—through empirical and experimental inquiry.

The conceptions of truth, inquiry, and reality proposed by the classic thinkers of the pragmatist tradition have engendered both realist and constructivist (and idealist) readings. On the one hand, Peirce, James, and Dewey advanced the view that reality is, in some sense, independent of our opinions of it. In Peirce's account of the scientific method of inquiry, that method is aimed at uncovering a reality that exists independently of our beliefs or perceptions (Rydenfelt 2021a). Nevertheless, the pragmatists also proposed various ways in which such a reality is malleable to human thought and understanding. Early in his career, Peirce (1871) suggested that while reality is independent of any particular thought, it is not independent of thought in general, positioning him close to what is often called objective idealism (see Lane 2017, chapter 3). Dewey, in turn, has often been read as a proponent of constructivism, which maintains that reality is, in some way, dependent on our interactions with it (Godfrey-Smith 2016). A central example of this constructivist dimension is Dewey's account of the “transactions” between the “knower” and the “known”, where both are transformed (Dewey & Bentley 1949a, b). Importantly, however, regardless of how we choose to interpret the pragmatists on this score—highlighting either realist or constructivist elements—the same interpretation applies

across all domains of inquiry. For the pragmatists, there is no categorical divide separating the subject matters studied by ethicists and, say, physical scientists: it is not suggested that the results of physics lend themselves more readily to a realist interpretation, while those of ethical inquiry allow for a more constructivist rendering.

The notion of pragmatism as a mediating position in ontology and epistemology is a recurring theme in James's 1906 lectures published as *Pragmatism* (James 1907), where James presented pragmatism as an alternative to two types of intellectual temperaments: the tender-minded, characterized by rationalism, idealism, optimism, and religiosity, and the tough-minded, defined by empiricism, materialism, pessimism, and irreligiosity. While tough-minded thinkers are more attuned to the actualities of human life and experience, their worldview offers little room for the ideals and the consolidation of religious notions. On the other hand, the tender-minded, turning to idealist or religious metaphysics, are disconnected from the realities of experience. James proposed “the oddly-named thing pragmatism as a philosophy that can satisfy both kinds of demand”, as it is “religious like the rationalisms” while retaining “the richest intimacy with facts” typical of empiricism (James 1907, 13). While James attempts to strike a balance between the ontological and epistemological commitments of the two intellectual temperaments, the significance of the resulting mediating pragmatist position is ethical. Despite their apparent differences, both tough-mindedness and tender-mindedness, driven by their respective pessimism and optimism, lead to a similar outcome: inaction with respect to the actualization of ideals. In contrast, pragmatism charts a middle path of meliorism that—again put in religious terms—“treats salvation as neither inevitable nor impossible”, but rather “as a possibility, which becomes more and more of a probability the more numerous the actual conditions of salvation become” (James 1907, 110).² Despite the religious language employed, James is clear that meliorism is not necessarily religious in nature; instead, it remains open to the ideals which “any one of us in this room” cherishes and “is willing to live and work for” (James 1907, 110). These ideals, meliorism maintains, are not mere abstractions but live possibilities that can be put into practice, achievable through sustained effort to reach the conditions for their actualization.

This notion of meliorism—along with a similarly critical diagnosis of optimism and pessimism—was adopted by Dewey. In his view, pessimism is a “paralyzing doctrine”

² Unlike fallibilism, meliorism is not a pragmatist coinage. The term was likely first used by the novelist George Eliot in the 1870s, while its contrast with optimism and pessimism was already articulated in the 1880s and 1890s by the botanist and sociologist Lester Ward (Bergman 2015). For a discussion on Peirce's complicated relationship with meliorism, see Bergman (2012; 2015)

that renders futile any effort to identify and address the remediable causes of specific evils, undermining attempts to improve the world, while optimism, explaining away such evils, is “equally an incubus” (Dewey 1920, 181–182). In contrast to James, however, Dewey promoted an activist conception of meliorism as a call for the concrete involvement of philosophers in addressing actual social problems (Bergman 2015). In line with his views on social inquiry and democracy, this entailed a new conception of social philosophy. Unlike the traditional philosopher, who remains confined to concepts and the relationships between ideas, the Deweyan social philosopher was to help improve human life by supplying “hypotheses to be used and tested in projects of reform” (Dewey 1920, 190). Dewey’s views, in turn, influenced and were influenced by thinkers such as Jane Addams, the social progressive often associated with the pragmatist tradition. Addams’s notion of lateral progress, rather than defining progress through its most outstanding achievements, emphasized the collective betterment of society, a process that is slower but broader in scope; this view has sometimes been described as “radical meliorism” (Hamington 2022).

4 The interplay of fallibilism and meliorism

The notions of fallibilism and meliorism allow to set up distinctions that clarify pragmatism’s position with respect to the two narratives Watson et al. distinguish in terms of the four premises derived from Danaher. Regarding the first and second premises—the fact premise and value premise—pragmatism offers a fallibilist epistemological stance that avoids the extremes of dogmatism and skepticism. It suggests that there are no principled barriers to learning the ways the world is, while also recognizing that there are no indubitable beliefs that would act as a permanent foundation for science. Inquiry proceeds through continual revision and correction, grounded in experience and experimentation. With respect to the third and fourth premises—the evaluation premise and technological premise—pragmatism proposes another mediating position, meliorism. While neither technological nor societal progress is inevitable, the human condition is susceptible to active improvement. The alternatives here are sociotechnical optimism and pessimism. Sociotechnical optimism accepts the latter two premises; sociotechnical pessimism denies one or both of them, contesting either “progress” or, at least, its connection to technological developments.

Moreover—beyond an exercise in conceptual clarification—these distinctions help illuminate potential issues and areas for improvement within the proposal for sociotechnical pragmatism. Watson et al. allude to pragmatism having epistemological and ethical dimensions; both are certainly

present in their discussion. However, Watson et al. primarily rely on either considerations motivated by either fallibilist or meliorist dimensions of pragmatism in their discussion, rather than fully engaging with the interplay of the two theses as pragmatism would propose. Maintaining a distinction between the two dimensions allows us to see how they must interact for pragmatism to provide a genuine alternative to sociotechnical dogmatism and skepticism, as well as sociotechnical optimism and pessimism.

In parts of Watson et al.’s discussion, epistemological considerations take center stage. This is most evident in their treatment of the debate surrounding the potential of explainable AI (XAI) to help humans understand and review the opaque behavior of complex statistical models, such as deep neural networks. Watson et al. argue that the dogmatist believes new and improved methods will eventually resolve issues of transparency. In contrast, they connect the skeptical narrative to the suspicion that XAI might offer misleading explanations, creating a false sense of understanding that could lead to faulty decisions and harmful consequences. As a third alternative, they propose a “pragmatic approach” to XAI that begins with an agent in context, acknowledging that XAI tools can inform decision-making under the right conditions (Watson et al. 2024, 14).

Watson et al. claim that the distinction between (what they call) dogmatism and skepticism “reflects the longstanding tension between realism and constructivism in epistemology” (Watson et al. 2024, 3). They associate sociotechnical dogmatism with a belief in the objectivity and singularity of truth, and attribute to sociotechnical skepticism a contrasting commitment to epistemological constructivism (or subjectivism), which holds that “true explanations” “cannot exist and would be unknowable if they did” (Watson et al. 2024, 14). However, framing the distinction between the two narratives in these terms is problematic. It aligns realism with the idea that the world is knowable and constructivism with the idea that it is not. The reverse is often true, however: realists may well be skeptical about the knowability of an “independent” reality, while constructivists argue that reality becomes knowable precisely because of the human element in its very constitution. The true tension lies not between realism and constructivism but between skepticism and dogmatism in an epistemological sense. As we have seen, pragmatism proposes fallibilism as a middle ground between these two extremes, maintaining that while there are no principled barriers to the scope of human inquiry and knowledge, none of our commitments should be regarded as beyond criticism and revision. It is because the original epistemological meanings of these terms have not been retained by Watson et al. that the pragmatist alternative remains unclear.

This problematic framing has significant consequences for both Watson et al.’s interpretation of pragmatism and its

potential relevance to the debate on the transparency of AI and the potential of XAI. Watson et al. position pragmatism as an alternative to both realism and constructivism, foregrounding a “radical” context-dependency of inquiry and the view that facts are “relational” rather than subjective or relative. As already noted in connection with the first premise, Watson et al. maintain that pragmatists accept that relevant facts can be assembled, yet argue that “explanations are radically context-dependent”—even suggesting that pragmatism holds truth to be “not as important as what is useful” (Watson et al. 2024, 15). However, while Watson et al. seek to distinguish their understanding of facts as “relational” from relativism, their emphasis on radical context dependence and the prioritization of what is useful gives rise to concerns and aligns uncomfortably closely with the skeptical narrative. Although pragmatism readily acknowledges that different kinds of explanations may prove useful in different contexts, shifting the focus from truth to usefulness—or, even more problematically, equating truth with usefulness—opens the door to the idea that explanations are relative to particular practical contexts, and may even be inconsistent with one another. When applied to the issue of explainability in machine learning and AI, such an account easily lends itself to the implication that our ability to capture crucial aspects of how AI systems function—in turn enabling us to audit and validate their operations and trace their potential significance—would hinge on whether we happen to pose the “right” questions or demand the “right” kinds of explanations.

While pragmatism stresses both holism and the piecemeal, evolving nature of inquiry, its fallibilism does not suggest that we are restricted to purely contextual inquiries and explanations, nor that facts are, in the relevant sense, relational. Although it acknowledges that inquiry is always provisional and unfinished, pragmatism nonetheless allows for the possibility of explanations that extend beyond immediate contextual constraints; however fallible those explanations may be. This clarification is central for understanding the alternative that pragmatism offers. Grounded in fallibilism, the significance of sociotechnical pragmatism in relation to AI lies in its contention that, contrary to the skeptical position, there are no principled barriers to explaining presently opaque machine learning models. At the same time, however, pragmatism recognizes that there are no guarantees that any limited number of inquiries—let alone the introduction of further technologies such as XAI—will conclusively resolve issues of explainability.

Another issue in the vicinity is that much of the relevance of sociotechnical pragmatism as a contrasting narrative, as Watson et al. develop it, lies in its argument that we can obtain greater insight into the consequences and implications of AI. Watson et al. claim that the risks associated with XAI are primarily “epistemic” rather than ethical, a view

that aligns with broader AI debates, where transparency has frequently been elevated to a value seemingly for its own sake (Jobin et al. 2019; see Munn 2023). Even more generally, Watson et al. suggest that sociotechnical pragmatism places procedural regularity and transparency at the core of good governance, valuing “justification through informed, rational, and honest discourse” (Watson et al. 2024, 10). However, the value of transparency is often instrumental rather than intrinsic—as a means to understanding the consequences and significance of AI systems and identifying errors. While the problems involved may most immediately be epistemic in nature, the risks usually extend beyond such issues alone. Reliance on fallibilism—even as elaborated here—should not lead to an exclusive focus on transparency and procedural viability while neglecting the substantive actionability of ethical ideals, resulting in fallibilism without meliorism.

To be sure, Watson, Mökander, and Floridi’s proposal does not lack what was above distinguished as the meliorist dimension of pragmatism. This is particularly evident in their proposal of sociotechnical pragmatism as advancing regulation aimed at fostering AI technologies for socially beneficial purposes while mitigating risks. For example, discussing their second central case, the fairness of AI systems, Watson et al. argue that a dogmatist narrative results in attempts to “automate our way out of fundamental social problems like systematic discrimination and structural inequality” (Watson et al. 2024, 11). Such an approach is not only problematic in itself but also overlooks the possibility that fairness could be achieved without relying solely on technology. In contrast, Watson et al.’s pragmatist perspective considers AI systems deployed in high-risk domains on a case-by-case basis, and advances good regulation that promotes both growth and pro-social outcomes.

Once again, the distinction between fallibilism and meliorism helps clarify the mediating role that pragmatism may offer. Watson et al. link restrictive regulation of technological systems to sociotechnical skepticism as one of its implications. On the other hand, as just seen, they associate sociotechnical pragmatism with regulatory efforts, making it difficult to distinguish their proposed position from sociotechnical skepticism. As already argued, however, while skepticism accurately describes the rejection of epistemological premises 1 and 2, the position contesting the third or (at least) the fourth premise is better characterized as pessimism. Such pessimism is not necessarily tied to regulatory efforts; rather, it suggests that technological systems do not improve the evaluation of future developments, even in the presence of regulatory interventions. In this Jamesian sense, rejecting the third or fourth premise contributes to inaction, as the negative outcome is perceived as inevitable. From this perspective, the contrast between the consequences of sociotechnical optimism and pessimism, on the one hand,

and pragmatism becomes clear: while both optimism and pessimism lead to inaction—either by assuming that technological progress will naturally resolve issues or the very opposite—sociotechnical pragmatism as a meliorist project aims at improvement by active measures, such as the regulatory efforts of shaping technological development Watson et al. propose.

This elaboration of the alternative provided by pragmatism also exposes how the melioristic aspect of Watson et al.'s proposal is structured around relatively fixed and predetermined ideals. In their account, sociotechnical pragmatists propose evaluating the design and use of AI systems in terms of how they “advance or hinder specific normative goals” requiring policymakers to be clear about the normative ends they aim to achieve (Watson et al. 2024, 8). While pragmatists, in this view, are open to experimentation and empirical determination of “what does and does not work,” they argue that such experimentation is ideally conducted “according to public and predetermined criteria” (Watson et al. 2024, 8). However, this risks conveying the impression that meliorism is founded on pre-established ends, which are then hoped to be enacted through experimentation and regulation. In turn, this renders Watson et al.'s proposal susceptible to what might be characterized as regulatory optimism: not only is regulation and policy-making seen as practically capable of shaping the future, but the normative foundations on which such efforts rest are also treated as readily available. Such an optimism overlooks the fallibilist dimension of pragmatism with respect to normative starting points.

Indeed, Watson et al. give relatively little direct attention to how sociotechnical pragmatism approaches the second, value premise. They suggest that “what counts as morally acceptable is a revisable cultural inheritance” and that there is “no reason to believe that this inheritance should offer theoretical closure,” concluding that “we should not expect normative tensions to be overcome” by technological innovation or regulation (Watson et al. 2024, 15). However, the formulation appears too strong, lending itself to the implication that such a prospect can be ruled out in advance—a stance that contrasts with fallibilism and borders on skepticism. More broadly, rather than simply focusing on historicity of our ethical perspectives, pragmatism advocates an empirical and experimental approach to values and norms. Our ethical commitments are to be subjected to inquiry, not merely used or applied as a guide in the process of evaluating and regulating technology. It is this interplay of fallibilism and meliorism that pragmatism may further contribute to debates on the ethical and societal significance of sociotechnical developments.

This contribution becomes especially relevant in light of an emerging body of research examining how values change in response to new technologies. Based on the notion that technologies can expand or constrain decision-making

options and enable new interpretations of moral situations (Verbeek 2011), recent research has proposed that the adoption and use of sociotechnical systems may lead to shifts in moral beliefs and practices (van de Poel 2021), including changes in central normative concepts such as truth and trust (Danaher & Sætra 2023). However, it is easy to assume that such value change arises from difficulties in “applying” existing ethical perspectives to new situations brought about by technological and sociotechnical transformations. Alternatively, value change may be understood as occurring when new technologies provide information or insights that alter how we perceive the world, prompting adjustments in our practices to better align with our values. As already suggested, however, pragmatism maintains that novel and problematic situations themselves can serve as catalysts for inquiry into the ethical theories and perspectives underlying our particular values, exposing their potential limitations and errors.

Indeed, a pragmatist approach to value change has recently been developed by Ibo van de Poel and Olya Kudina (2022). Drawing on Deweyan concepts, they propose that technology-induced value change can occur through the reinterpretation of values in a specific situation (which they term value dynamism), the subsequent modification of value generalizations that influence future contexts (value adaptation), or the emergence of entirely new values from indeterminate situations (value emergence). However, even here there remains further room to explore the full potential of a pragmatist and fallibilist perspective in this context. While van de Poel and Kudina (2022) reference Dewey's experimental approach to value change, their focus is primarily on how values have already changed or may continue to do so in response to new, indeterminate situations. Less attention is given to how this revision of values could itself be understood as, or actively shaped into, a deliberative process of inquiry, akin to the employment of experimental methods in other domains of research. A fallibilist stance on our normative starting points—including established ethical values, norms, and principles—can serve as a counterbalance to potential overconfidence in the activity prompted by meliorism, including regulatory efforts. The need for such a counterbalance becomes particularly relevant when considering the practical differences between Watson et al.'s proposal and pragmatism viewed as sociotechnical fallibilism and meliorism in light of a central counterargument.

5 Pragmatism and the logic of theodicy

Watson, Mökander, and Floridi's proposal for sociotechnical pragmatism involves both fallibilist and meliorist elements that are central to philosophical pragmatism. The foregoing discussion has highlighted how keeping these dimensions

distinct enables us to see how pragmatism offers an alternative to both skeptical and dogmatist (in the epistemological sense) and optimistic and pessimistic (in the ethical sense) approaches to the development of new technologies. These distinctions also serve to illuminate the interplay of these elements in the pragmatist approach, showing how the fallibilist element encourages inquiry into the values, norms, and ideals that underlie melioristic activity. To further clarify the practical difference between Watson et al.'s proposal and the conception of pragmatism as sociotechnical fallibilism and meliorism, we may now turn to a final consideration. With these distinctions in hand, pragmatism can be shown capable of addressing a challenge that could be levied against Watson et al.'s proposal as an approach to sociotechnical issues.

This challenge comes from a direction that may seem unexpected, namely, the issue of theodicy and its recent criticism. The so-called problem of theodicy is typically addressed in theology and the philosophy of religion. Since antiquity, philosophers and theologians have pondered how an omnipotent, omniscient, and benevolent God could allow horrendous evil and suffering in the world. Traditional theodicies have sought to offer reasons for why God permits such evil. However, it has recently been proposed that such a “theodacist logic” is not confined to religious contexts but can be identified in a variety of non-religious settings—including political decision-making, administration, social practices, and public institutions (Pihlström 2020). The unifying feature of this theodacist logic is the effort to justify, excuse or find meaning in the suffering of others by appealing to a supposed “common good” that the suffering allegedly serves. To combat this pattern and reject the theodacist presuppositions that underpin secular practices and institutions, an *anti-theodacist* perspective that eschews such justifications and excuses has been proposed for renewing ethics within contemporary societies (Pihlström 2022).

At the first glance, the abstract criticism of theodacist logic may seem distant from the issues at hand. To grasp the relevance of theodacist logic in the context of sociotechnical development and AI, consider an alternative narrative that expresses sociotechnical optimism and pessimism in practical terms—one that foregrounds a tension between two centers of power seeking to shape and govern the development of digital technologies. This duality has been evident in discussions on the regulation of social media platforms and data-driven services (Tambini 2021; Schroeder 2025), and features prominently in contemporary public discourse on AI (MacCarthy & Propp 2021; Siegmann & Anderljung 2022; AlgorithmWatch 2024). In this narrative, one power node—often called “Silicon Valley”—of giant tech companies rapidly develops and pushes new products and services to the market, empowered by a “pragmatic” attitude of moving forward, seeing what happens, only later expressing regret or paying (often disproportionately small) fines

if necessary. The other power node—call it “Brussels”—consists of administrations, bureaucrats, policymakers, and politicians attempting to devise new forms of regulation for these technologies amid a deluge of white papers, policy proposals, and lobbying efforts. While the relative influence of these two forces has varied over time, the story goes, Silicon Valley appears to have the upper hand with a renewed push for deregulation and more relaxed regulation. This trend is reinforced by the present U.S. administration's emphasis on deregulation, particularly under the banner of protecting freedom of speech (Echikson 2025). The rhetoric of freedom and deregulation, however, may serve as a pretext for further consolidating control over political and societal discourse, as well as for expanding the extraction of user data—especially within platforms owned by powerful actors closely aligned with the administration.

This narrative, while certainly a rough sketch, enables us to see how a pessimistic vision of our sociotechnical future—or a form of sociotechnical pessimism—can build upon various observations. Silicon Valley, driven by economic interests, displays little concern for the negative impacts of their technologies on the human condition, as documented in critical research. Moreover, its operations are difficult to constrain by the means at the disposal of Brussels—insignificant sanctions and slow, cumbersome regulatory procedures often trailing far behind the availability of new technologies. Even worse, Brussels has been infiltrated with representatives of Silicon Valley, shaping regulatory efforts with their economic influence, with business leaders and executives expressing concerns over the implications of regulation on innovation, economic growth, and global competitiveness. It is a short step to the pessimistic conclusion that we live in the midst of a human and societal experiment of an unprecedented scale. New technologies, including AI, are being introduced at an unforeseen pace, with the documented potential for excessively harmful consequences, including actual human suffering. Meanwhile, following a theodacist logic, these developments are justified by claims about the potential benefits of these technologies in terms of economic growth, scientific breakthroughs, and overall improvement to the human condition.

A similar anti-theodacist criticism might be well raised in connection with Watson, Mökander, and Floridi's mediating narrative of sociotechnical pragmatism that, by and large, proposes that we observe how technologies develop, assess their consequences, and regulate them accordingly. This narrative is based on the view that we may successfully investigate the operations of new technologies and their effects on human life, while recognizing that our actions may—but do not necessarily—lead to improvements in the world. In practice, however, this position risks becoming indistinguishable from a “wait and see” approach, grounded in the confidence that emerging technologies can be understood, reviewed,

and assessed as they unfold and that potential harms can concomitantly be addressed through regulation.

Nevertheless, also the vision of pragmatism as a combination of sociotechnical fallibilism and meliorism, as developed here, could be subject to a similar criticism. This pragmatist perspective suggests that inquiry should engage not only with the actualized and expected repercussions of sociotechnical developments but also with their unintended consequences, including the revision of values prompted by problematic situations. Does this not similarly propose a vast societal experiment where even devastating consequences for individuals and, in the worst case, all of humanity, might be explained and justified as necessary costs in the advancement of inquiry?

Despite the structural similarities of these objections, however, the differences between the two alternative pragmatist approaches points to a significant difference in practice. Watson et al.'s proposal largely rests on two central elements. The first is the rejection of both sociotechnical skepticism and dogmatism, grounded in the belief that we can understand how new technologies function—whether through explainable AI or other methods—and assess their consequences. The second is the assumption that regulation and policy-making can shape deployment of new technologies to improve outcomes and prevent worst-case scenarios. As we have seen, in practice, the combination of these views reflects a form of regulatory optimism. In the first instance, understanding how technologies work is expected to take place during or after their introduction. In the second, Watson et al. largely presuppose that the values guiding regulation are reasonably well established, providing a foundation for shaping the design, introduction, and use of emerging technologies.

Pragmatism as sociotechnical fallibilism and meliorism, as developed here, offers a different vision of approaching technological developments. Much of our assessment of the consequences and implications of new technologies will inevitably take place after their introduction into actual human practices. However, experimentation need not unfold on the grand scale of entire societies or humanity. Pragmatism instead encourages proactive inquiry, anticipating potential consequences before large-scale implementation. Moreover, extending fallibilism to normative questions about the values and principles that should guide technological development, this approach resists the assumption that we must wait to see how values shift in response to technological change. Instead, it suggests that we can anticipate how emerging situations may call for revisions to the values that currently shape regulation and governance. As Watson et al. (2024, 8) nicely put it, pragmatism views emancipation as “neither inevitable nor final”. What provides the possibility of such emancipation, however, is a commitment to research that does not simply react to technological change

but actively engages in anticipating and critically examining its implications before they materialize. Pragmatism does not attempt to justify or explain away potential harms as necessary costs of progress; it resists a theodictist logic by insisting that the ethical and societal implications of emerging technologies are actively interrogated and adjusted before large-scale deployment by means of controlled processes of inquiry.

Indeed, by providing an alternative vision of democratic governance of AI technologies, pragmatism as sociotechnical fallibilism and meliorism provides a basis for challenging the overall condition in which power over technological development is concentrated in the hands of a few key actors—such as Silicon Valley and Brussels of our earlier narrative. Recent research serves to help illustrate this alternative. Researchers have increasingly highlighted the democratic implications of AI. For example, Taylor et al. (2024) argue that democratizing AI is essential for fostering trust, equity, and fairness, particularly in the deployment of AI-enabled technologies within public administration, and advocate for a Maximum Feasible Participation (MFP) approach to AI development. However, how such democratization should be implemented, and how participatory elements should be incorporated into AI systems, remains largely unresolved. Instead, in line with the narrative just discussed, AI development is largely driven by technical experts and large corporations, while regulation is similarly conducted by a relatively small group of political decision-makers. As suggested by a recent study by Hedlund and Persson (2025) examining the perspectives of various experts—including technical developers, policymakers, and other specialists—forward-looking responsibility in AI development is primarily understood in terms of “contribution” that is closely linked to an actor’s ability to effect change in AI development, with influence frequently cited by study participants as a key criterion for responsibility.

Sociotechnical pragmatism, informed by the principles of fallibilism and meliorism, presents an alternative vision. First, adopting this approach would require extending both the examination of AI and decision-making beyond a limited number of experts—such as technological developers, major corporations, and policymakers—on the basis that the consequences of technological development cannot be fully anticipated from any number of restricted perspectives. Instead, the assessment of emerging technologies and of policies concerning their regulation should be distributed across multiple actors and institutions capable of contributing to empirical inquiry by identifying risks, evaluating consequences, and revising normative assumptions accordingly. Second, this broader distribution of processes of reviewing and making decisions concerning new technologies aligns with Dewey’s pragmatist conception of democracy as a form of social inquiry, where democratic participation

is not confined to formal governance but involves collective problem-solving through empirical means. This view acknowledges the indispensability of expertise in addressing complex societal challenges that cannot be resolved solely through public discussion and deliberation. However, while processes of social inquiry may often be expert-driven, they also involve affected individuals and communities as active contributors. Applying this view of democracy to AI governance would entail treating affected individuals and communities not merely as passive recipients of new technologies or policies but as sources of information and data on problems, potential solutions, and the observed consequences of new technologies and the policies regulating them. Advancing such an approach to AI governance as social inquiry would introduce a form of democracy that would break the narrative of technological development as dictated solely by industry priorities or static regulatory norms.

6 Conclusion

This discussion has sought to refine and extend Watson, Mökander, and Floridi's timely proposal for sociotechnical pragmatism by clarifying its relationship to the broader pragmatist tradition, particularly through the notions of fallibilism and meliorism. While Watson et al.'s proposal incorporates both fallibilist and meliorist dimensions, the way they frame the opposition between alternative narratives introduces conceptual ambiguities that complicate the articulation of pragmatism as a distinct position. Their distinction between sociotechnical dogmatism and sociotechnical skepticism conflates epistemological and ethical dimensions, making it difficult to separate the proposed pragmatist stance from both the inclination toward regulation that they associate with skepticism and the regulatory optimism that their confidence in the capacity to examine and govern technology appears to entail.

The pragmatist alternative in the study of technology, it has been argued, is best understood through the interplay of fallibilism and meliorism. Fallibilism offers an epistemological stance that neither assumes definitive knowledge of the function of new technologies and their repercussions, nor denies the possibility of understanding them, as exemplified by its stance toward the potential of explainable AI. Meliorism, in turn, introduces an ethical orientation toward improvement without assuming that progress is inevitable. Together, these two dimensions distinguish pragmatism from varieties of both sociotechnical dogmatism and skepticism, on the one hand, and sociotechnical optimism and pessimism, on the other. Pragmatism does not presuppose that technological development will necessarily lead to progress, nor does it withdraw into skepticism or inaction in response to uncertainty. Instead, it treats inquiry as

an ongoing, iterative process of understanding and shaping technological change, including the potential revisions to values and norms that may emerge in response to the new situations introduced by technological developments.

In the context of AI in particular, a central practical difference follows in relation to the regulation of these technologies. Watson et al.'s proposal tends toward regulatory optimism, assuming that governance mechanisms can direct technological deployment in socially beneficial ways. However, this assumes that the normative foundations of regulation are already established, and risks making policy interventions reactive rather than proactive. By contrast, the interplay of fallibilism and meliorism resists the assumption that values are fixed and instead treats them as subject to inquiry. Rather than assuming that governance should adapt to technological change as it unfolds, pragmatism suggests that ethical inquiry should be integrated into the development process itself, ensuring that potential consequences and normative shifts are actively examined in advance. By doing so, pragmatism as sociotechnical fallibilism and meliorism proposes a way to avoid the risks of a great human and societal experimentation which treats harm as an inevitable cost of progress. Moreover, the pragmatist conception of democracy as social inquiry offers an alternative vision of AI governance where the assessment of emerging technologies is distributed across multiple actors and institutions, and affected individuals and communities are incorporated as active contributors rather than passive recipients of new technologies and policies.

Acknowledgements I am indebted to David Hildebrand, Brendan Hogan, Sami Pihlström, and Samuel Singler for the comments and discussions.

Author contributions This article is by a single author.

Funding Open Access funding provided by University of Helsinki (including Helsinki University Central Hospital). This work was funded by Research Council of Finland, 361684, Koneen Säätiö, Helsingin Sanomain Säätiö.

Data availability No datasets were generated or analyzed during the current study.

Declarations

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will

need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- AlgorithmWatch. (2024). Europe's approach to AI regulation: Embracing Big Tech and security hardliners. <https://algorithmwatch.org/en/ai-regulation-europe-big-tech-and-security/>
- Bergman M (2012) Improving our habits: peirce and meliorism. In: De Waal C, Skowronski KP (eds) *The normative thought of Charles S. Peirce*. Fordham University Press, New York, pp 125–148
- Bergman, M. (2015). Minimal meliorism: Finding a balance between conservative and progressive pragmatism. In *Action, Belief and Inquiry: Pragmatist Perspectives on Science, Society and Religion* (pp. 2–28). Nordic Pragmatism Network.
- Bohman J (2010) Participation through publics: did Dewey answer Lippmann? *Contemp Pragmatism* 7(1):49–68
- Danaher J (2022) Techno-optimism: an analysis, an evaluation and a modest defence. *Philos Technol*. <https://doi.org/10.1007/s13347-022-00550-2>
- Danaher J, Sætra HS (2023) Mechanisms of techno-moral change: a taxonomy and overview. *Ethical Theory Moral Pract* 26(5):763–784. <https://doi.org/10.1007/s10677-023-10397-x>
- Dewey J (1920) Reconstruction in philosophy. In: Boydston JA (ed) *The middle works of John Dewey, 1899–1924*, vol 12. Southern Illinois University Press, Carbondale, pp 77–201
- Dewey J (1927) The public and its problems. In: Boydston JA (ed) *The later works of John Dewey, 1925–1953*, vol 2. Southern Illinois University Press, Carbondale, pp 235–372
- Dewey J (1929) The quest for certainty: a study of the relation of knowledge and action. In: Boydston JA (ed) *The later works of John Dewey, 1925–1953*, vol 4. Southern Illinois University Press, Carbondale, pp 1–255
- Dewey J (1935) Liberalism and social action. In: Boydston JA (ed) *The later works of John Dewey, 1925–1953*, vol 11. Southern Illinois University Press, Carbondale, pp 1–65
- Dewey J (1938) Logic: the theory of inquiry. In: Boydston JA (ed) *The later works of John Dewey, 1925–1953*, vol 12. Southern Illinois University Press, Carbondale, pp 1–527
- Dewey J, Bentley AF (1949a) *Knowing and the known*. Beacon Press, Boston
- Dewey J, Bentley AF (1949b) *Knowing and the known*. In: Boydston JA (ed) *The later works of John Dewey, 1925–1953*, vol 16. Southern Illinois University Press, Carbondale, pp 1–294
- Echikson W (2025) Trump, tech and transatlantic turbulence. *Eur View*. <https://doi.org/10.1177/17816858251339298>
- Godfrey-Smith P (2016) Dewey and the question of realism. *Noûs* 50(1):73–89. <https://doi.org/10.1111/nous.12059>
- Hamington, M. (2022). Jane Addams. *The Stanford Encyclopedia of Philosophy* (Fall 2022 edition). <https://plato.stanford.edu/archives/fall2022/entries/addams-jane/>
- Hedlund M, Persson E (2025) Distribution of responsibility for AI development: expert views. *AI Soc*. <https://doi.org/10.1007/s00146-024-02167-9>
- James W (1907) *Pragmatism: a new name for some old ways of thinking*. Longmans, Green and Co, New York
- James W (1909) *The meaning of truth*. Longman Green and Co., New York
- Jobin A, Ienca M, Vayena E (2019) The global landscape of AI ethics guidelines. *Nat Mach Intell* 1(9):389–399. <https://doi.org/10.1038/s42256-019-0088-2>
- Lane R (2017) *Peirce on realism and idealism*. Cambridge University Press, Cambridge
- MacCarthy, M., & Propp, K. (2021). *Machines learn that Brussels writes the rules: The EU's new AI regulation*. Brookings Institution. <https://www.brookings.edu/articles/machines-learn-that-brussels-writes-the-rules-the-eus-new-ai-regulation/>
- Munn L (2023) The uselessness of AI ethics. *AI Ethics* 3(3):869–877. <https://doi.org/10.1007/s43681-022-00209-w>
- Peirce CS (1871) Fraser's the works of George Berkeley. In: Houser N, Kloesel C (eds) *The essential peirce: selected philosophical writings (1867–1893)*, vol 1. Indiana University Press, Bloomington, pp 83–105
- Peirce CS (1878) How to make our ideas clear. In: Houser N, Kloesel C (eds) *The essential peirce: selected philosophical writings (1867–1893)*, vol 1. Indiana University Press, Bloomington, pp 124–141
- Peirce CS (1903) The three normative sciences. In: Peirce CS (ed) *The essential peirce: selected philosophical writings (1893–1913)*, vol 2. Indiana University Press, Bloomington, pp 196–207
- Pihlström S (2020) Theodicy by other means?: Rethinking “god after auschwitz” through the dialectics of antitheodicism. *Cosmos Hist* 16(2):475–494
- Pihlström S (2022) Beyond the theory-practice dichotomy: pragmatism, antitheodicy, and the recognition of suffering. In: Hirvonen O, Koskinen HJ (eds) *The theory and practice of recognition*. Routledge, London, pp 305–322
- Putnam H (1992) A reconsideration of deweyan democracy. In: Putnam H (ed) *Renewing philosophy*. Harvard University Press, Cambridge, pp 180–202
- Russell B (1910) Pragmatism. In: Russell B (ed) *Philosophical essays*. Routledge, London, pp 79–111
- Rydenfelt H (2013) Constructivist problems, realist solutions. In: Kegley J, Skowronski K (eds) *Persuasion and compulsion in democracy*. Lexington Books, Lanham, pp 153–169
- Rydenfelt H (2019) Pragmatism, social inquiry and the method of democracy. In: Holma K, Kontinen T (eds) *Practices of citizenship in East Africa: perspectives from philosophical pragmatism*. Routledge, London, pp 29–43
- Rydenfelt H (2021a) Realism without representationalism. *Synthese* 198(4):2901–2918. <https://doi.org/10.1007/s11229-019-02251-4>
- Rydenfelt H (2021b) Recent problems of the public. In: Festl M (ed) *Pragmatism and social philosophy*. Routledge, London, pp 237–247
- Rydenfelt H (2022a) Pragmatism and analytic philosophy. In: Aikin S, Talisse R (eds) *Routledge companion to pragmatism*. Routledge, London, pp 138–146
- Rydenfelt H (2022b) Environmental pragmatism and the revision of values. *Trans Charles S. Peirce Soc* 58(1):52–66. <https://doi.org/10.2979/trancharpeirsoc.58.1.04>
- Rydenfelt H (2023a) Pragmatism and experimental bioethics. *Cambridge Q Healthc Ethics* 33(2):174–184. <https://doi.org/10.1017/S0963180123000282>
- Rydenfelt H (2023b) Controversial views and moral realism. *Theor Med Bioeth* 44(2):165–176. <https://doi.org/10.1007/s11017-023-09616-4>
- Rydenfelt H (2024) Promises and pitfalls of environmental pragmatism. *Environ Values* 33(4):380–393. <https://doi.org/10.1177/09632719231196542>
- Rydenfelt H (2025) Inquiring value: the pragmatist turn in business ethics. *J Bus Ethics* 197:1–11. <https://doi.org/10.1007/s10551-024-05711-1>
- Schroeder R (2025) Content moderation and the digital transformations of gatekeeping. *Policy Internet* 17(2):e425
- Siegmann, R., & Anderljung, M. (2022). *Machines learn that Brussels writes the rules: The EU's new AI regulation and its global impact*. Brookings Institution. <https://www.brookings.edu/artic>

- [les/machines-learn-that-brussels-writes-the-rules-the-eus-new-ai-regulation/](#)
- Tambini D (2021) Media freedom. Polity Press, Cambridge
- Taylor RR, Murphy JW, Hoston WT et al (2024) Democratizing AI in public administration: improving equity through maximum feasible participation. *AI Soc.* <https://doi.org/10.1007/s00146-024-02120-w>
- Van de Poel I (2021) Design for value change. *Ethics Inf Technol* 23(1):27–31. <https://doi.org/10.1007/s10676-018-9461-9>
- van de Poel I, Kudina O (2022) Understanding technology-induced value change: a pragmatist proposal. *Philos Technol* 35(2):40. <https://doi.org/10.1007/s13347-022-00520-8>
- Verbeek PP (2011) *Moralizing technology: understanding and designing the morality of things*. University of Chicago Press, Chicago
- Watson DS, Mökander J, Floridi L (2024) Competing narratives in AI ethics: a defense of sociotechnical pragmatism. *AI Soc.* <https://doi.org/10.1007/s00146-024-02128-2>
- Publisher's Note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.